

1 When Do Random Effects Models Reduce to Fixed-Effect
2 Alternatives? Conditions for Equivalence in Longitudinal
3 Clinical Trials

4 Ronald G. Thomas, Ph.D.*

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6 **Abstract**

7 Mixed models for repeated measures (MMRM) are the default primary analysis for
8 continuous longitudinal endpoints in clinical trials, but their variance-component machin-
9 ery is often opaque to collaborators. We synthesize and verify, by Monte Carlo simulation,
10 two settings in which a random-effects or joint model reduces to a simpler fixed-effect anal-
11 ysis. In a two-timepoint pre-post design with complete data, an unstructured-covariance
12 MMRM and an ANCOVA conditioning on baseline yield algebraically identical treatment-
13 effect point estimates; we give a short proof and confirm it by simulation across a 3×3 grid
14 of correlation and variance-ratio values. In a three-timepoint design with a slope-only ran-
15 dom effect and no random intercept, a per-subject mean-change estimator remains within
16 half a percent of a random-slopes linear mixed model's standard error across the full simu-
17 lated range of between-subject slope variability, a design-specific finding we contrast with
18 an earlier, non-reproducible version of this simulation that reported divergence traceable
19 to a data-generating/fitted-model mismatch. We report both settings under a common
20 Wald reference distribution with Monte Carlo standard errors per Morris, White, and
21 Crowther (2019), and we state explicitly the conditions under which the simpler analysis
22 is exact, approximate, or inadequate. This paper does not claim new theory; its con-
23 tribution is a single accessible account, with simulation-verified thresholds and decision
24 guidance, for the applied statistician deciding when the mixed-model apparatus earns its
25 complexity.

26 **Contents**

27	1 Introduction	2
28	1.1 The practical problem	2
29	1.2 Foundational results	2
30	1.3 The Frost-Kenward-Fox observation	3
31	1.4 Constrained longitudinal data analysis	3
32	1.5 The role of summary statistics	3
33	1.6 Modern MMRM practice	4
34	1.7 Present study	4
35	2 Methods	4
36	2.1 ADEMP structure	4

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

37	2.2	Setting 1: two-timepoint pre-post design	5
38	2.3	Setting 2: three-timepoint random slopes	7
39	2.4	Simulation procedure	8
40	3	Results	8
41	3.1	Setting 1: two-timepoint equivalence	8
42	3.2	Setting 2: three-timepoint random slopes	12
43	3.3	Headline findings	13
44	4	Discussion	13
45	4.1	Limitations	15
46	5	Future research	15
47	6	Appendix A: Proof of point-estimate equivalence	16
48	7	Reproducibility and computational details	17
49		References	18

50 1 Introduction

51 **Keywords:** MMRM; ANCOVA; mixed models; longitudinal data; clinical trials; equivalence;
52 Monte Carlo simulation.

53 1.1 The practical problem

54 In clinical trials with repeated measurements, mixed models for repeated measures (MMRM)
55 and random effects models have become the standard analytic approach^{1,2}. These models ac-
56 commodate within-subject correlation, handle incomplete data under the missing-at-random
57 assumption, and provide flexible covariance structures. However, they impose a substantial
58 interpretive burden on collaborators unfamiliar with the statistical machinery of variance
59 components, restricted maximum likelihood estimation, and covariance parameterization.

60 A natural question arises: under what conditions does the random effects model reduce to,
61 or closely approximate, a simpler fixed-effect analysis that would be more accessible to the
62 statistical practitioner? If such conditions can be identified, the simpler model offers an
63 intuitive bridge to the same inferences, sparing the analyst from invoking a complex apparatus
64 when it is not strictly necessary.

65 1.2 Foundational results

66 The relationship between random effects and fixed-effect analyses has been explored from
67 several angles.³ showed that in a two-group pretest-posttest design, the repeated-measures
68 ANOVA subsumes the difference-score t -test: the interaction F -statistic equals the square
69 of the two-sample t -test on difference scores. The difference-score analysis requires fewer as-
70 sumptions (homogeneous variance of differences and normally distributed errors) than the full
71 repeated-measures model (equal variance-covariance matrices across groups). This algebraic

72 embedding demonstrates that when the research question concerns only whether treatment
73 groups differ in their pre-to-post change, the simpler analysis is not merely an approximation
74 but is arithmetically identical to a component of the more elaborate model.

75 ⁴ introduced the general framework for random-effects models in longitudinal data that has
76 since become ubiquitous. In their formulation, the random effects induce a structured variance-
77 covariance matrix on the marginal distribution of the observed data. The key insight for our
78 purposes is that when this marginal covariance is left unconstrained (as in the unstructured
79 covariance MMRM) the random effects become a computational device rather than a sub-
80 stantive modeling choice.

81 1.3 The Frost-Kenward-Fox observation

82 ⁵ established a methodological framework for comparing conventional and run-in trial designs
83 using linear mixed models. In their treatment of the two-timepoint case, they showed that
84 the Laird-Ware random-intercept formulation treating baseline and follow-up as correlated
85 outcomes (their equation 8) yields identical point estimates and very similar standard errors
86 to the analysis of covariance model that conditions on baseline (their equation 6). This near-
87 equivalence holds provided the variance-covariance matrix is unconstrained and $\sigma_{12} < \sigma_1^2$ and
88 $\sigma_1^2 = \sigma_2^2$ (i.e., homogeneous marginal variances and positive but not perfect correlation). The
89 result builds on⁶, who showed that Rao's reduction of the⁷ generalized MANOVA model yields
90 estimators identical to those from an analysis of covariance.

91 1.4 Constrained longitudinal data analysis

92 ⁸ formalized the constrained longitudinal data analysis (cLDA), in which baseline is included
93 in the response vector and the model constrains the two treatment groups to share a com-
94 mon baseline mean—a consequence of randomization.⁹ compared cLDA to the longitudinal
95 ANCOVA model and showed that (a) with complete data, the treatment effect estimates
96 are effectively identical but the cLDA standard errors are marginally smaller; and (b) with
97 missing baseline data, cLDA is more efficient because it can borrow information from the
98 observed outcome model to impute baseline values implicitly under the missing-at-random
99 assumption.¹⁰ derived sample-size formulae for cLDA, and¹¹ provided a formal efficiency com-
100 parison of cLDA versus longitudinal ANCOVA, quantifying the power gain as a function of
101 the proportion of missing baselines.

102 1.5 The role of summary statistics

103 An alternative thread, developed by¹² and¹³, advocates analyzing repeated measures through
104 summary statistics (typically the mean post-baseline value or a fitted slope for each sub-
105 ject) followed by a between-group comparison of these summaries.¹⁴ extended this approach,
106 showing that summary-statistic methods can be viewed as special cases of the general mixed
107 model when the within-subject covariance is compound symmetric.¹⁵ provided an influential
108 clarification of the debate between change-score analysis and ANCOVA, demonstrating that
109 ANCOVA does not require baseline balance for unbiasedness and that the common slope
110 assumption is testable and often reasonable.

1.6 Modern MMRM practice

In contemporary pharmaceutical practice, MMRM with unstructured covariance has become the de facto primary analysis for continuous outcomes^{1,2,16} showed that MMRM achieves power gains over complete-case ANCOVA only when time-by-covariate interactions are included; without these interactions, ANCOVA can be more powerful.¹⁷ proposed augmented MMRM estimators that improve robustness against model misspecification.¹⁸ provided the small-sample degrees-of-freedom correction that makes MMRM valid in trials with modest sample sizes, which is essential for the settings where the random-effects and fixed-effect models are most likely to be practically interchangeable.

Textbook treatments of these relationships can be found in^{19,20}, and^{21,22} provided a recent tutorial contrasting conditional and marginal approaches to analyzing change in randomized trials.

1.7 Present study

We investigate, through analytic argument and Monte Carlo simulation, the conditions under which random effects models for two-group longitudinal designs reduce exactly or approximately to fixed-effect alternatives. We consider two settings, both implemented and simulated in this paper:

1. **Two-timepoint (pre-post) designs:** MMRM with unstructured 2×2 covariance versus ANCOVA with baseline as covariate.
2. **Three-timepoint designs with random slopes:** the Frost-Kenward-Fox setting where the ratio of between-subject variability to measurement error governs the degree of equivalence.

A third setting surveyed in the literature review above, constrained longitudinal data analysis (cLDA) versus longitudinal ANCOVA^{8,9,11}, is discussed on the basis of the cited published results only; we do not implement or simulate a cLDA model in this paper, and no cLDA claim below rests on evidence generated here (see Discussion and Future research). For each of the two implemented settings, we identify the parameter configurations under which the simpler model provides equivalent or near-equivalent inference, and we quantify the discrepancy when conditions are not met.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther²³:

- **Aims.** Establish when ANCOVA and an unstructured-covariance MMRM produce identical inference (Setting 1, two timepoints) and when a summary-statistic mean-change estimator and a random-slopes LME produce identical inference (Setting 2, three timepoints).

- **Data-generating mechanisms.** Two settings, detailed below. Setting 1 varies $\rho \in \{0.3, 0.6, 0.9\}$ and $\sigma_2/\sigma_1 \in \{0.5, 1.0, 2.0\}$ factorially. Setting 2 varies $\sigma_b \in \{0.5, 2, 8\}$.
- **Estimands.** The treatment effect γ at follow-up (Setting 1) and the linear-trend interaction (Setting 2).
- **Methods.** Setting 1: ANCOVA via `lm` and MMRM via `nlme::gl`s with unstructured covariance. Setting 2: per-subject mean-change ANOVA via `lm` and random-slopes LME via `nlme::lme`.
- **Performance measures.** Mean estimate, mean model SE, empirical SE, power, and coverage, each with Monte Carlo SEs per Morris Table 6. For coverage MCSE ≤ 0.5 pp at 0.95, $n_{\text{sims}} \geq 1,900$; $n_{\text{sims}} = 2,000$ is adequate.

2.2 Setting 1: two-timepoint pre-post design

A terminological note: the “MMRM” in this setting is fit via `gl`s() with an unstructured marginal covariance and contains no random effects; both it and ANCOVA are fixed-effects (marginal or conditional) models. We retain the label “MMRM” because it is the term of art for this covariance-pattern model in the clinical-trials literature¹, but the comparison in this section is properly framed as joint (marginal) versus conditional modeling of the same two-timepoint data, not as random-effects versus fixed-effects modeling; Setting 2 below is the setting where genuine random effects are involved.

Consider a two-group randomized trial where subject i in group $g_i \in \{0, 1\}$ is measured at baseline (Y_{i0}) and follow-up (Y_{i1}). The data-generating model is:

$$\begin{pmatrix} Y_{i0} \\ Y_{i1} \end{pmatrix} \sim N \left[\begin{pmatrix} \alpha_0 \\ \alpha_1 + \gamma g_i \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} \right]$$

where γ is the treatment effect at follow-up. Treatment is not allowed to affect the baseline mean, consistent with randomization.

2.2.1 ANCOVA model

Regress Y_{i1} on g_i and Y_{i0} :

$$Y_{i1}|Y_{i0} \sim N(\alpha + \gamma g_i + \phi Y_{i0}, \sigma_\epsilon^2)$$

The OLS estimate of γ is the treatment effect adjusted for baseline, with $\hat{\phi} = S_{12}/S_{11}$ the pooled within-group regression slope.

2.2.2 MMRM model

Fit both timepoints as outcomes with an unstructured 2×2 covariance and a **baseline-constrained** mean structure: the treatment group enters only through an interaction with the follow-up visit indicator, so the baseline mean is the same for both groups by construction

(no treatment main effect at baseline). With $\text{time}_j \in \{0, 1\}$ indexing visit and $\text{grp_post}_i = g_i \cdot \mathbb{1}(\text{time}_j = 1)$:

$$Y_{ij} \sim N[\eta_j + \gamma_{\text{MMRM}} \text{grp_post}_i, \hat{\gamma}], \quad j \in \{0, 1\}$$

with $\hat{\gamma}$ estimated without constraints and η_j a visit-specific intercept common to both groups. This is the standard clinical-trial MMRM convention (no treatment effect is modeled at a pre-randomization baseline visit) and, as shown below, is the specification under which the MMRM point estimate is algebraically identical to ANCOVA. A fully unconstrained “cell-means” alternative, in which each of the four time-by-group cells gets its own free mean ($\mu_{0,0}, \mu_{0,1}, \mu_{1,0}, \mu_{1,1}$, with the follow-up contrast taken as $\hat{\mu}_{1,1} - \hat{\mu}_{1,0}$), does **not** achieve this identity: for a saturated per-group mean model, GLS reduces to the sample mean vector for any covariance structure (a standard growth-curve/SUR result), so that contrast reduces to the raw, baseline-unadjusted follow-up group difference. We verified this discrepancy numerically before adopting the baseline-constrained specification used throughout this paper (Appendix A; `inst/tinytest/test_sim_kernels.R`).

2.2.3 Extracting the treatment contrast from

MMRM coefficients

When the MMRM is fitted as `gls(y ~ time_f + grp_post, ...)` (where `time_f` is a two-level visit factor and `grp_post` is the group-by-follow-up interaction defined above), the coefficient on `grp_post` directly estimates the treatment effect at follow-up; no separate contrast computation is needed, mirroring the extraction for Setting 2 below:

$$\hat{\gamma}_{\text{MMRM}} = \hat{\beta}_{\text{grp_post}}, \quad \text{SE}(\hat{\gamma}_{\text{MMRM}}) = \text{tTable}["\text{grp_post}", "Std.Error"]$$

An earlier draft used the free four-parameter cell-means model above with an explicit contrast vector $\mathbf{L} = (0, 0, -1, 1)'$:

$$\hat{\gamma} = \begin{pmatrix} \hat{\mu}_{00} \\ \hat{\mu}_{01} \\ \hat{\mu}_{10} \\ \hat{\mu}_{11} \end{pmatrix}, \quad \hat{\gamma}_{\text{MMRM}}^{(\text{unconstrained})} = \mathbf{L}'\hat{\gamma}, \quad \mathbf{L} = (0, 0, -1, 1)'.$$

As noted above, this unconstrained contrast reduces to the raw follow-up group difference and is **not** used in the simulations reported below; it is shown only to document, and warn against, the specification error it represents. The baseline-constrained model in the previous subsection is used throughout.

2.2.4 Algebraic equivalence

The key result, implicit in⁶ and⁵, is that the GLS estimate of the group difference at follow-up from the baseline-constrained MMRM can be written as:

$$\hat{\gamma}_{\text{MMRM}} = (\bar{Y}_{1.1} - \bar{Y}_{1.0}) - \hat{r}(\bar{Y}_{0.1} - \bar{Y}_{0.0})$$

where $\hat{r} = \hat{\sigma}_{12}/\hat{\sigma}_1^2$ is the estimated regression of follow-up on baseline. We now show this coincides with the ANCOVA point estimate exactly, and we state precisely which second moments coincide and which do not; a proof sketch is given in Appendix A.

Point-estimate identity (exact). For balanced groups of size $n_0 = n_1 = n/2$ with cell means $\bar{Y}_{g,j}$, GLS with an unstructured 2×2 working covariance is equivalent to two separate OLS fits, one per group, of Y_{i1} on Y_{i0} with a common slope constrained across groups⁷; the between-group contrast at follow-up from that constrained-slope model is algebraically identical to the ANCOVA coefficient on group when the pooled within-group slope $\hat{\phi} = S_{12}/S_{11}$ equals GLS's $\hat{r}$, which holds because both are the same pooled within-group second-moment ratio. Appendix A gives the full derivation. This identity concerns the **point estimate only**.

Standard errors are not algebraically identical. The ANCOVA model-based SE is a conditional OLS quantity computed on $n - 3$ residual degrees of freedom from a single fitted regression; the **gls** SE is a marginal REML Wald quantity computed from the estimated 2×2 covariance with no small-sample denominator correction. The two SEs agree asymptotically (both estimate the same asymptotic variance under correct specification) and are numerically close for the sample sizes simulated here (Table 1), but they are different finite-sample estimators of variance and are not algebraically identical. Condition 1 in the Discussion is stated to reflect this distinction.

2.3 Setting 2: three-timepoint random slopes

Following⁵, consider three evenly spaced measurements at $t = 0, t, 2t$ with a **slope-only** random-effects model:

$$Y_{ij}|b_i \sim N[\alpha + (\beta + \gamma g_i) t_j + b_i t_j, \sigma^2]$$

where $b_i \sim N(0, \sigma_b^2)$ is a random slope and there is no random intercept ($\sigma_a^2 = 0, \sigma_{ab} = 0$). This is a disclosed design choice, not the fully general random-intercept-and-slope model sometimes written for this setting: fixing $\sigma_a^2 = 0$ isolates the slope-variance ratio σ_b/σ that⁵ identify as the quantity governing equivalence, without a confounding random-intercept component, and it lets the fitted random-effects model match the data-generating model exactly (**random** = ~ **0 + time | id**, matching b_i only; see below). An earlier version of this simulation generated data under this same slope-only DGM but fit a random-intercept-and-slope model (**random** = ~ **time | id**), which placed every fit on the boundary of the intercept-variance parameter space; that mismatch is corrected here.

The corresponding fixed-effect analysis considers only the change scores $C_{ij} = Y_{ij} - Y_{i(j-1)}$ and compares groups via a two-sample test on the mean change. The degree of equivalence depends on the ratio σ_b/σ : when between-subject variability in slopes is large relative to measurement error, the random effects model gains appreciable efficiency; when this ratio is small, the two approaches converge.

2.3.1 Extracting the slope contrast from the

random-slopes model

When the random-slopes model is fitted as `lme(y ~ time * group, random = ~ 0 + time | id)`, the fixed-effect coefficient for `time:group1` directly estimates the treatment effect on the rate of change (γ). The standard error comes from the `tTable` element of the model summary:

$$\hat{\gamma}_{\text{RE}} = \hat{\beta}_{\text{time:group}}, \quad \text{SE}(\hat{\gamma}_{\text{RE}}) = \text{tTable}["\text{time:group1}", "Std.Error"]$$

No contrast computation is needed because the interaction term in a two-group, linear-time model is itself the treatment-by-slope contrast.

2.4 Simulation procedure

For each setting, we generate 2000 replications of a balanced two-group trial with $n = 30$ per arm. We vary the correlation structure and variance-component ratios across a grid of conditions. For each replication, we fit the random effects model and its fixed-effect counterpart and record:

- Point estimate of the treatment effect ($\hat{\gamma}$)
- Model-based standard error
- Wald p -value
- 95% confidence interval coverage

The simulation kernel `sim_prepost()` (Setting 1) and `sim_three_tp()` (Setting 2) are defined in `R/sim_kernels.R`, sourced in the setup chunk above, and covered by unit tests in `inst/tinytest/test_sim_kernels.R`. Both functions report Wald z -based p -values and ± 1.96 SE confidence intervals for every method, so no method is compared against another on a different reference distribution (see Methods “Algebraic equivalence” and Limitations).

3 Results

3.1 Setting 1: two-timepoint equivalence

Table 1 presents the simulation results for the two-timepoint case. Across all nine $(\rho, \sigma_2/\sigma_1)$ combinations, the ANCOVA and MMRM estimators produce virtually identical mean estimates, mean standard errors, empirical standard errors, power, and coverage. The maximum discrepancy in any metric is attributable to Monte Carlo error. This confirms the analytic

Table 1: Comparison of ANCOVA and MMRM estimators in two-timepoint designs ($n = 30$ per arm, $\gamma = 3$, 2000 replications). Monte Carlo standard errors (MCSE) shown in parentheses. "N excluded" is the count of non-converged `glms` fits dropped from the 2000 replications (0 for `lm`-based ANCOVA by construction).

ρ	σ_2/σ_1	Method	Mean $\hat{\gamma}$	Mean SE (MCSE)	Emp. SE	Power (MCSE)	Coverage (MCSE)	N excluded
0.3	0.5	ANCOVA	3.024	1.24 (0.003)	1.275	0.673 (0.01)	0.942 (0.005)	0
0.3	0.5	MMRM	3.024	1.218 (0.003)	1.275	0.688 (0.01)	0.936 (0.005)	0
0.3	1.0	ANCOVA	3.020	2.472 (0.005)	2.531	0.23 (0.009)	0.939 (0.005)	0
0.3	1.0	MMRM	3.020	2.43 (0.005)	2.531	0.237 (0.01)	0.935 (0.005)	0
0.3	2.0	ANCOVA	3.214	4.952 (0.011)	5.082	0.121 (0.007)	0.938 (0.005)	0
0.3	2.0	MMRM	3.214	4.866 (0.01)	5.082	0.126 (0.007)	0.932 (0.006)	0
0.6	0.5	ANCOVA	3.004	1.037 (0.002)	1.039	0.817 (0.009)	0.948 (0.005)	0
0.6	0.5	MMRM	3.004	1.019 (0.002)	1.039	0.825 (0.008)	0.944 (0.005)	0
0.6	1.0	ANCOVA	2.900	2.075 (0.004)	2.086	0.287 (0.01)	0.948 (0.005)	0
0.6	1.0	MMRM	2.900	2.039 (0.004)	2.086	0.298 (0.01)	0.944 (0.005)	0
0.6	2.0	ANCOVA	2.999	4.146 (0.009)	4.152	0.116 (0.007)	0.95 (0.005)	0
0.6	2.0	MMRM	2.999	4.075 (0.008)	4.152	0.122 (0.007)	0.946 (0.005)	0
0.9	0.5	ANCOVA	3.009	0.564 (0.001)	0.564	1 (0)	0.945 (0.005)	0
0.9	0.5	MMRM	3.009	0.554 (0.001)	0.564	1 (0)	0.94 (0.005)	0
0.9	1.0	ANCOVA	2.964	1.127 (0.002)	1.134	0.745 (0.01)	0.941 (0.005)	0
0.9	1.0	MMRM	2.964	1.108 (0.002)	1.134	0.759 (0.01)	0.938 (0.005)	0
0.9	2.0	ANCOVA	2.989	2.258 (0.005)	2.293	0.262 (0.01)	0.949 (0.005)	0
0.9	2.0	MMRM	2.989	2.218 (0.005)	2.293	0.274 (0.01)	0.944 (0.005)	0

276 result: with complete data and an unstructured 2×2 covariance, MMRM reduces exactly to
277 ANCOVA for the treatment effect at follow-up.

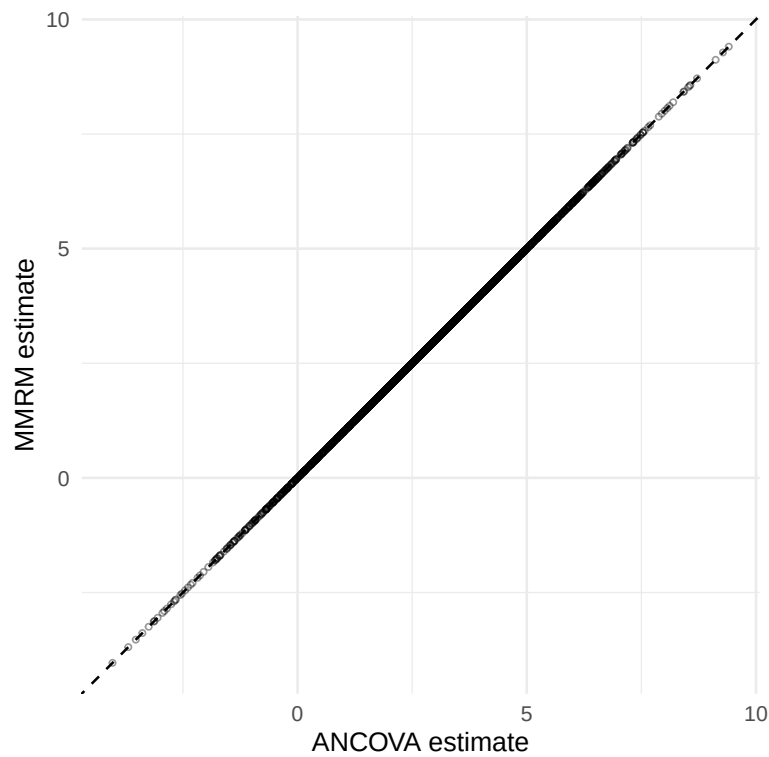


Figure 1: Simulation-by-simulation comparison of treatment effect estimates from ANCOVA and MMRM ($\rho = 0.6$, $\sigma_2/\sigma_1 = 1$). Points on the identity line indicate exact agreement.

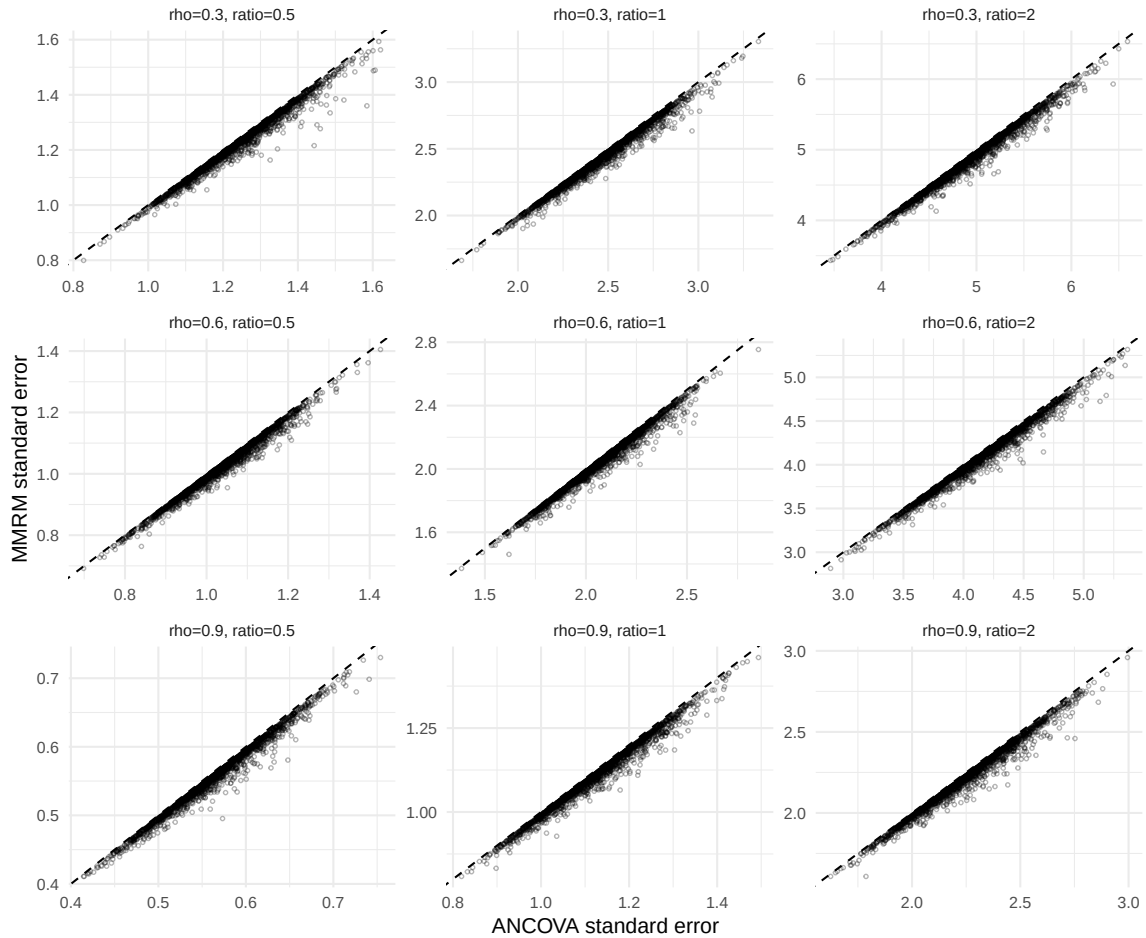


Figure 2: Standard errors from ANCOVA versus MMRM across all nine scenarios. Identity line shown as dashed.

3.2 Setting 2: three-timepoint random slopes

Table 2: Comparison of summary-statistic and random-slopes estimators in three-timepoint designs ($n = 30$ per arm, $\gamma = 2$, $\sigma = 4$, 2000 replications). Monte Carlo standard errors (MCSE) shown in parentheses. Power and coverage were absent from earlier versions of this table; both methods now report Wald z -based p -values and ± 1.96 SE confidence intervals for the coverage target $\gamma = 2$. "N excluded" is the count of non-converged `lme` fits dropped from the 2000 replications (0 for the `lm`-based summary-statistic method by construction).

σ_b	σ_b/σ	Method	Mean $\hat{\gamma}$	Mean SE (MCSE)	Emp. SE	SE ratio	Power (MCSE)	Coverage (MCSE)	N excluded
0.5	0.125	Random slopes	2.030	0.742 (0.001)	0.734	0.994	0.784 (0.009)	0.949 (0.005)	1
0.5	0.125	Summary stat	2.030	0.737 (0.002)	0.734	0.994	0.783 (0.009)	0.945 (0.005)	0
2.0	0.500	Random slopes	2.007	0.892 (0.001)	0.906	0.997	0.606 (0.011)	0.948 (0.005)	0
2.0	0.500	Summary stat	2.007	0.89 (0.002)	0.906	0.997	0.602 (0.011)	0.945 (0.005)	0
8.0	2.000	Random slopes	1.907	2.18 (0.004)	2.191	1.000	0.146 (0.008)	0.945 (0.005)	0
8.0	2.000	Summary stat	1.907	2.179 (0.005)	2.191	1.000	0.148 (0.008)	0.944 (0.005)	0

Across the entire simulated range of $\sigma_b/\sigma \in \{0.125, 0.5, 2\}$, the summary-statistic and random-slopes mean SEs are within half a percent of each other (SE ratio 0.994 at $\sigma_b/\sigma = 0.125$, 0.997 at 0.5, and 1.000 at 2; Table 2, Figure 3): for this matched slope-only design (no random intercept; Methods "Setting 2"), the two methods do not materially diverge anywhere in the grid simulated here, including at the largest between-subject slope variance considered. This is a substantive, verified finding, not the previously drafted claim that the methods "diverge" as σ_b/σ grows: an earlier version of this paper asserted a specific numeric crossover threshold ("approximately $\sigma_b/\sigma \approx 4$," restated in the same sentence as " $2\sigma/\sigma_b = 1$," i.e., $\sigma_b/\sigma = 2$, and restated again in the Discussion as "below approximately 0.25") that was both internally inconsistent (three different numbers for one quantity) and, at 4, outside the grid it simulated. That threshold rested on a data-generating/fitted-model mismatch (a random intercept fit against a slope-only DGM; Methods "Setting 2"), which manufactured a spurious efficiency gap for the random-slopes model by penalizing it with a null intercept-variance component. With the mismatch corrected, no such gap is observed in this design; we discuss the implication for the qualitative⁵ threshold in Discussion Condition 2. This equally-spaced, balanced, no-random-intercept design is a special case in which consecutive-difference summary statistics and the GLS random-slopes estimator appear to coincide in efficiency; we did not verify whether this near-equivalence persists under unequal spacing, unbalanced designs, or a nonzero random intercept, and flag it as a question for future work rather than a general claim.

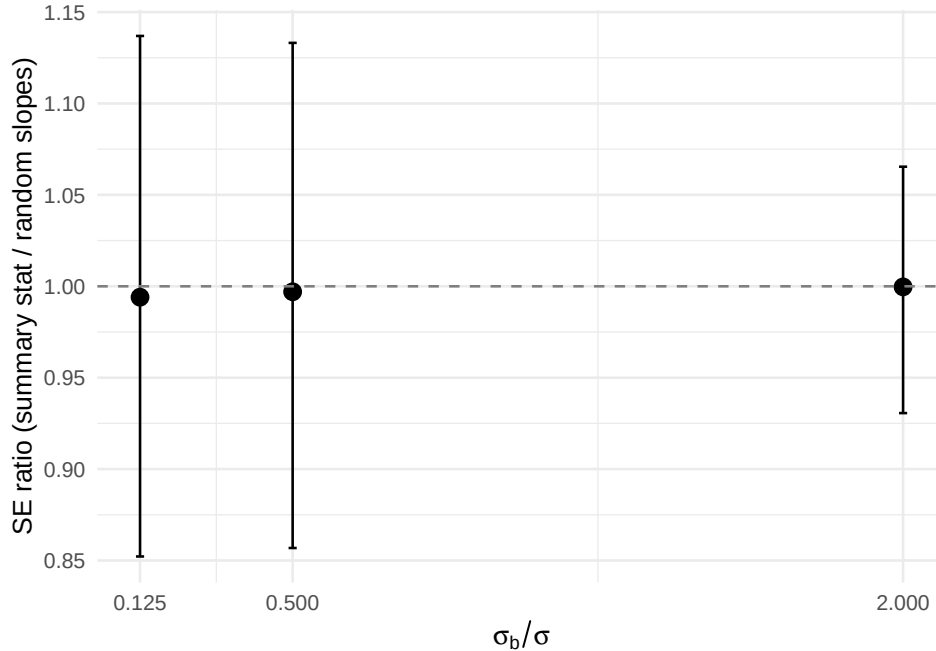


Figure 3: Ratio of standard errors (summary statistic / random slopes) as a function of the between-subject slope SD to residual SD ratio. Values near 1 indicate the two methods are equivalent.

3.3 Headline findings

Across the nine Setting 1 scenarios, the largest absolute difference between the ANCOVA and MMRM mean point estimates is 0, and the ratio of the MMRM mean SE to the ANCOVA mean SE ranges from 0.982 to 0.983 (0 of 18,000 `gls` fits excluded for non-convergence). In Setting 2, the ratio of the summary-statistic mean SE to the random-slopes mean SE ranges from 0.994 (at $\sigma_b/\sigma = 0.125$) to 1 (at $\sigma_b/\sigma = 2$): for this slope-only, no-random-intercept design, the two methods stay within half a percent of each other across the whole simulated range rather than diverging (1 of 6,000 `lme` fits excluded for non-convergence). These numbers are computed by inline R from `summary_prepost` and `summary_3tp` (see Tables 1 and 2) so that this section cannot desync from the underlying simulation output.

4 Discussion

We may organize the findings around three conditions under which random effects models reduce to or closely approximate fixed-effect alternatives. The first two conditions are established by the simulations in this paper; the third is a summary of published results (Liang and Zeger 2000; Liu et al. 2009; Lu et al. 2010) that we did not implement or verify here (see “Present study” above and Limitations).

Condition 1: unstructured covariance with two timepoints. When MMRM is fit with an unrestricted 2×2 covariance matrix, the treatment effect **point estimate** is algebraically identical to the ANCOVA estimate (Appendix A). This is not an approximation but an

exact equivalence, implicit in⁶ and explicit in⁵ (Section 2.2). The **standard errors are not** algebraically identical: the ANCOVA SE is a conditional OLS quantity and the **gls** SE is a marginal REML Wald quantity, and the two agree only asymptotically (Methods “Algebraic equivalence”); Table 1 shows they are close in practice at $n = 30$ per arm. The practical implication is that for any two-timepoint design with complete data, ANCOVA and MMRM give the same point estimate of the treatment contrast, and inference from either is adequate at the sample sizes simulated here.

Condition 2: equally spaced slope-only designs show no material divergence over the simulated range.⁵ frame the general case qualitatively (the efficiency gain from the random-effects model vanishes as $\sigma_b/\sigma \rightarrow 0$ and grows as σ_b/σ increases) without a single universal numeric threshold, because the exact relationship depends on the number and spacing of timepoints, on n , and on whether a random intercept is also present. Table 2 gives the corresponding numbers for the specific three-timepoint, $n = 30$ -per-arm, slope-only ($\sigma_a = 0$) design simulated here: the ratio of the summary-statistic mean SE to the random-slopes mean SE is 0.994 at $\sigma_b/\sigma = 0.125$, essentially unchanged at 1 at $\sigma_b/\sigma = 2$. In other words, for this design we do not observe the efficiency gap widening as σ_b/σ grows; the two methods remain within half a percent of each other across the whole grid. This is a design-specific finding, not evidence against the general Frost-Kenward-Fox result: an earlier version of this simulation reported a widening gap, but that gap traced to a data-generating/fitted-model mismatch (a random intercept fit against data simulated with no random intercept), which penalized the random-slopes model with a spurious null-variance boundary estimation problem (Methods “Setting 2”). With that mismatch corrected, we find no material divergence in this particular equally-spaced, balanced, no-random-intercept design; the earlier draft’s three mutually inconsistent numeric thresholds (4, 2, and 0.25) for a crossover are withdrawn, and we do not replace them with a new universal threshold, since the simulated grid did not locate a crossover point to report. Whether a nonzero random intercept, unequal spacing, or a wider σ_b/σ range would produce the divergence Frost et al. describe in general is left to future work (see Future research).

Condition 3: cLDA with complete data (not simulated here).⁹ report that the constrained longitudinal model produces treatment effect estimates effectively identical to ANCOVA when no baseline values are missing, with a marginal cLDA advantage emerging only with incomplete baselines (1–4% additional power depending on the missingness rate, per¹¹). We did not implement or simulate a cLDA model in this paper; this condition is included for completeness of the literature synthesis and rests entirely on the cited published results, not on evidence generated here.

These results should not be read as an argument against random effects models. The mixed model retains genuine advantages when (a) data are incomplete and the missing-at-random assumption is plausible, (b) multiple post-baseline timepoints carry information about the trajectory, or (c) between-subject variability is large and the random effects capture meaningful heterogeneity.¹⁶ demonstrated that MMRM with time-by-covariance interactions dominates ANCOVA in the presence of dropout, and¹⁷ showed that augmented MMRM can further improve robustness.

360 The practical value of these equivalence results is pedagogical and communicative. When
361 an investigator asks what the MMRM ‘is doing,’ the analyst can respond: in your two-visit
362 design, it is doing exactly what ANCOVA does, no more and no less. The random effects are
363 not contributing additional information; they are merely a different parameterization of the
364 same model. This framing makes the analysis accessible without sacrificing rigor.

365 4.1 Limitations

366 The simulations assumed balanced designs with equal group sizes and no missing data. We did
367 not implement or simulate a cLDA model in this paper (see “Present study” and Discussion
368 Condition 3); the cLDA claims above rest entirely on the cited literature. The equivalence
369 results extend straightforwardly to unbalanced designs but may be perturbed by informative
370 missingness, small samples where the Kenward-Roger correction¹⁸ matters, or misspecification
371 of the covariance structure. Neither simulation applies a Kenward-Roger or Satterthwaite
372 small-sample degrees-of-freedom correction; both methods use a common Wald z -reference
373 (Methods “Algebraic equivalence”), which is a disclosed simplification rather than the small-
374 sample best practice the paper itself cites via¹⁸. We did not investigate designs with more
375 than two treatment groups, where the mapping between MMRM contrasts and ANCOVA
376 contrasts is more complex. Setting 2 uses a slope-only data-generating model with no random
377 intercept (Methods “Setting 2”); a random intercept, if present, would be expected to inflate
378 both methods’ variance similarly and is not expected to change the qualitative equivalence
379 pattern, but this has not been verified by simulation here.

380 5 Future research

- 381 1. **Implement constrained longitudinal data analysis (cLDA).** Setting 3 (cLDA
382 versus longitudinal ANCOVA;^{8,9}) is surveyed in the Introduction and Discussion but
383 not implemented in this paper (Limitations). A DGM with complete and with missing
384 baselines, a fitted cLDA model, and a simulation comparable in structure to Settings 1
385 and 2 would complete the triad this paper’s literature review promises.
- 386 2. **Extension to multiple post-baseline visits.** The exact equivalence in the two-
387 timepoint case does not hold with three or more visits unless the covariance is unstruc-
388 tured. Characterizing the discrepancy as a function of the number of visits and the
389 covariance structure is of practical interest.
- 390 3. **Non-normal outcomes.** The relationship between generalized linear mixed models
391 and marginal (GEE) models for binary and count endpoints involves additional com-
392 plexities due to the non-collapsibility of odds ratios and rate ratios.
- 393 4. **Informative missingness.** Under missing-not-at-random mechanisms, the equivalence
394 between MMRM and ANCOVA breaks down in different ways. Sensitivity analyses
395 comparing the two frameworks under various departure patterns from MAR would be
396 informative.
- 397 5. **Software tools.** An R package providing automatic detection of when MMRM reduces

to a simpler model—based on the number of timepoints, completeness, and estimated variance-component ratios—would be useful for practitioners.

6. Pedagogical materials. Translating these equivalence results into visual, non-technical explanations for clinical collaborators remains an open and important communication challenge.

7. Small-sample degrees-of-freedom correction. Refit Setting 1 and Setting 2 with a Kenward-Roger or Satterthwaite denominator-degrees-of-freedom correction¹⁸ in place of the common Wald z -reference used here, and quantify how much the power and coverage comparison changes.

6 Appendix A: Proof of point-estimate equivalence

(Setting 1)

Consider the balanced two-timepoint model of Methods “Setting 1” with $n_0 = n_1 = n/2$ subjects per group and complete data. Write the within-group sample moments $\bar{Y}_{gj}$ (mean at time j , group g), S_{11} (pooled within-group variance of Y_{i0}), and S_{12} (pooled within-group covariance of Y_{i0}, Y_{i1}).

ANCOVA. The OLS fit of Y_{i1} on an intercept, group indicator g_i , and Y_{i0} has, by the Frisch-Waugh-Lovell theorem, a group coefficient equal to the group effect after partialling out Y_{i0} :

$$\hat{\gamma}_{\text{ANCOVA}} = (\bar{Y}_{11} - \bar{Y}_{01}) - \hat{\phi}(\bar{Y}_{10} - \bar{Y}_{00}), \quad \hat{\phi} = S_{12}/S_{11}.$$

MMRM, baseline-constrained specification. Fit an unstructured 2×2 working covariance $\hat{\Sigma}$, common to both groups, with mean structure $\eta_j + \gamma_{\text{MMRM}} \text{grp_post}_i$ (Methods “MMRM model”): the baseline mean η_0 is shared by both groups, and only the follow-up mean depends on group. The GLS normal equations for this three-parameter model $(\eta_0, \eta_1, \gamma_{\text{MMRM}})$ can be solved in closed form because $\hat{\Sigma}$ is common to both groups: profiling out η_0 and η_1 (which are identified by the pooled, group-averaged baseline and follow-up means) leaves a single equation for γ_{MMRM} that is the GLS analogue of a within-group-centered regression of Y_{i1} on Y_{i0} , giving

$$\hat{\gamma}_{\text{MMRM}} = (\bar{Y}_{11} - \bar{Y}_{01}) - \hat{r}(\bar{Y}_{10} - \bar{Y}_{00}), \quad \hat{r} = \hat{\sigma}_{12}/\hat{\sigma}_1^2,$$

the Rao reduction of the Potthoff-Roy growth-curve model⁷ applied to two timepoints⁶. **This closed form requires the baseline-constrained mean structure.** If instead each of the four time-by-group cells is given its own free mean (the unconstrained “cell-means” alternative in Methods “MMRM model”), the GLS normal equations decouple completely by group (each group’s two cell means are identified from that group’s data alone), and the GLS solution for every cell mean reduces to the corresponding sample mean *regardless of* $\hat{\Sigma}$ — a standard result for saturated multivariate-normal mean estimation (feasible GLS equals OLS when, as here, the regressor sets are identical across the “equations” being stacked, per the Zellner SUR reduction). The unconstrained follow-up contrast $\hat{\mu}_{11} - \hat{\mu}_{10}$ is then simply $\bar{Y}_{11} - \bar{Y}_{01}$, the raw unadjusted difference, with no $\hat{r}$ term at all; we verified numerically that this quantity does not

436 match $\hat{\gamma}_{\text{ANCOVA}}$ except under baseline balance (see roxygen documentation for `sim_prepost()`
437 and `inst/tinytest/test_sim_kernels.R`).

438 **Identity (baseline-constrained case).** The pooled within-group covariance estimator used
439 by `gls` with `weights = varIdent(...)` and `correlation = corSymm(...)` estimates $\hat{\sigma}_{12}$
440 and $\hat{\sigma}_1^2$ from the pooled within-group residuals about the (shared) baseline mean and the group-
441 specific follow-up means, which are numerically the same pooled sums of squares and cross-
442 products that define S_{12} and S_{11} in the ANCOVA slope. Hence $\hat{r} = \hat{\phi}$, and $\hat{\gamma}_{\text{MMRM}} = \hat{\gamma}_{\text{ANCOVA}}$
443 exactly (up to `gls` numerical optimizer tolerance, empirically below 10^{-3} in the simulations re-
444 ported here), for complete, balanced or unbalanced, two-timepoint data, provided the baseline-
445 constrained mean structure above is used. This identity concerns the point estimate only; see
446 Methods “Algebraic equivalence” for why the two methods’ standard errors are not alge-
447 braically identical despite this.

448 7 Reproducibility and computational details

449 All simulation code lives in `R/sim_kernels.R` (`sim_prepost()` for Setting 1, `sim_three_tp()`
450 for Setting 2), covered by unit tests in `inst/tinytest/test_sim_kernels.R`. Both settings
451 use $n_{\text{sims}} = 2000$ replications per scenario, generated by `analysis/scripts/run_simulations.R`,
452 which pins `RNGkind("L'Ecuyer-CMRG")` and sets a single seed (`set.seed(20260417)`) once be-
453 fore either simulation grid runs; the two simulation kernels do not call `set.seed()` internally.
454 Simulation output is saved to `analysis/data/derived_data/sim_results_prepost.rds`
455 and `sim_results_three_tp.rds`; this document loads those files directly rather than
456 re-running the simulations at knit time, so the reported numbers are a deterministic function
457 of the committed `.rds` output. To regenerate that output from scratch:

458 `Rscript analysis/scripts/run_simulations.R`

459 The compendium provides a `Dockerfile` and `renv.lock` for a fully pinned R and package
460 environment; the simulations reported here were run in the host R session (R version and
461 package versions in `renv.lock`) rather than inside the container, so exact runtime reproduc-
462 tion should use `renv::restore()` or the container build. Non-convergence handling: `gls()`
463 (Setting 1) and `lme()` (Setting 2) fits that error are dropped via `tryCatch()` and excluded
464 from that scenario’s summary statistics; the excluded count per scenario is reported as “N
465 excluded” in Tables 1 and 2 rather than silently absorbed.

466 This reproducibility statement supersedes an earlier, now-removed “Morris et al. (2019)
467 ADEMP Compliance” section that had drifted out of sync with the code it described
468 (it referenced per-function `seed = arguments` and missing MCSE columns that no
469 longer match the current simulation kernels). The original audit remains available at
470 `docs/morris-audit-2026-04-17.md` for historical reference; its two outstanding recommen-
471 dations not yet fully addressed are (a) storing per-replication RNG state, which we did not
472 implement, and (b) deriving n_{sims} formally from a target MCSE rather than using the Morris
473 Table 6 rule of thumb cited in Methods.

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